

Assignment 1 - EE4725 Quasi Optical Systems

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1 Theory: Far-Field Evaluation Using the Spectral Green's Function

The electric field radiated by an electric current distribution can be written as a spectral-domain convolution:

$$\vec{e}(x, y, z) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{\bar{G}}^{ej}(k_x, k_y) \vec{J}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} e^{-jk_z |z-z'|} dk_x dk_y \quad (1)$$

where $\bar{\bar{G}}^{ej}$ is the dyadic spectral Green's function (SGF) for electric currents and $\vec{J}$ is the Fourier transform of the current distribution.

By evaluating the integral at the stationary phase point, the far field is obtained as:

$$\vec{E}^{far}(r, \theta, \phi) = jk_{zs} \bar{\bar{G}}^{ej}(k_{xs}, k_{ys}, z, z') \vec{J}(k_{xs}, k_{ys}) \frac{e^{-jkr}}{2\pi r} \quad (2)$$

where the stationary phase point corresponds to:

$$k_{xs} = k \sin \theta \cos \phi, \quad k_{ys} = k \sin \theta \sin \phi, \quad k_{zs} = k \cos \theta \quad (3)$$

The dyadic SGF for electric currents in free space is:

$$\bar{\bar{G}}^{ej}(k_x, k_y) = -\frac{\zeta}{2k k_z} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x (\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y (\pm k_z) \\ -k_x (\pm k_z) & -k_y (\pm k_z) & k^2 - k_z^2 \end{bmatrix} \quad (4)$$

where $+k_z$ is used for $z > z'$ and $-k_z$ for $z < z'$. The spectral variable k_z is defined uniquely as:

$$k_z = -j \sqrt{-(k^2 - k_x^2 - k_y^2)} \quad (5)$$

The directivity is defined as:

$$D(\theta, \phi) = \frac{U(\theta, \phi)}{P_{\text{rad}}/(4\pi)} \quad (6)$$

where the radiation intensity is:

$$U(\theta, \phi) = \frac{|\vec{E}^{far}(r, \theta, \phi)|^2}{2\zeta} r^2 \quad (7)$$

and the total radiated power is:

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi U(\theta, \phi) \sin \theta d\theta d\phi \quad (8)$$

2 MATLAB Implementation

Four reusable MATLAB functions were implemented:

1. `EJ_SGF(er, k, kx, ky)` – computes the nine components of the spectral dyadic Green’s function $\bar{\bar{G}}^{ej}$.
2. `FTCurrent(k0, er, kx, ky, 1, w)` – computes the analytical Fourier transform of the dipole current distribution.
3. `farfield(k0, R_FF, TH, PH, kz, Gxx, Gyx, Gzx, Jx)` – evaluates the far field via the stationary phase formula (2). The product $jk_{zs}\bar{\bar{G}}^{ej}$ is formed by multiplying the passed SGF components with k_z ; at $\theta = \pi/2$ where $k_z \rightarrow 0$ and the SGF diverges, the resulting $0 \times \infty$ is replaced by the known analytical limit $-j\zeta/(2k)$ times the corresponding matrix element.
4. `Directivity(E_tot, Theta, dth, dph, er, r)` – computes the directivity and total radiated power by numerical integration.

3 Q1: Elementary Electric Source (3 points)

3.1 Problem Setup

An $\hat{x}$ -oriented elementary electric source is placed at the origin of the coordinate system:

$$\vec{j}(x, y, z) = \delta(x) \delta(y) \delta(z) \hat{x} \quad (9)$$

Its Fourier transform is simply $\vec{J}(k_x, k_y) = \hat{x}$.

Since the current is x -directed, only the first column of the dyadic SGF is relevant:

$$\vec{E} = \bar{\bar{G}}^{ej} \cdot \hat{x} = \begin{bmatrix} G_{xx} \\ G_{yx} \\ G_{zx} \end{bmatrix} \quad (10)$$

The parameters used are $f = 30$ GHz, $k_x \in [0, 3k_0]$, and $k_y = 0$.

3.2 Q1.1: SGF Components

Figure 1 shows the real and imaginary parts of the three SGF components G_{xx} , G_{yx} , and G_{zx} as a function of k_x/k_0 .

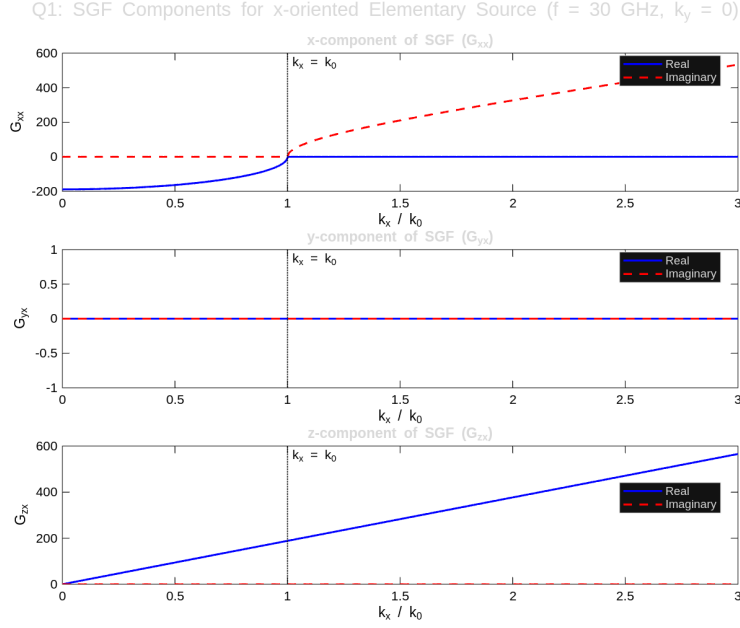


Figure 1: Real and imaginary parts of the SGF components G_{xx} , G_{yx} , and G_{zx} for an $\hat{x}$ -oriented elementary source at $f = 30$ GHz, $k_y = 0$. The dashed vertical line marks $k_x = k_0$.

Key observations from the plots:

- G_{xx} : real and negative for $k_x < k_0$, purely imaginary and positive for $k_x > k_0$. The singularity at $k_x = k_0$ corresponds to $k_z = 0$.
- G_{yx} : identically zero for all k_x because $k_y = 0$, so the off-diagonal term $-k_x k_y = 0$.
- G_{zx} : purely real over the entire range. This is because the k_z in the numerator ($-k_x k_z$) cancels the k_z in the denominator of the prefactor, giving $G_{zx} = \zeta k_x / (2k)$, which is real regardless of whether k_z is real or imaginary.

3.3 Q1.2: Imaginary Region of the x -Component

The x -component of the SGF is:

$$G_{xx} = -\frac{\zeta}{2k k_z} (k^2 - k_x^2) \quad (11)$$

- **Propagating region** ($k_x < k_0$, i.e. $k_x^2 + k_y^2 < k^2$): $k_z = \sqrt{k^2 - k_x^2}$ is real and positive. The prefactor $-\zeta/(2kk_z)$ is real, and $(k^2 - k_x^2)$ is real and positive. Therefore G_{xx} is **purely real** (and negative).
- **Evanescient region** ($k_x > k_0$, i.e. $k_x^2 + k_y^2 > k^2$): $k_z = -j\sqrt{k_x^2 - k^2}$ is purely imaginary. The prefactor becomes $-\zeta/(2k \cdot (-j|\cdot|))$, which is purely imaginary. Since $(k^2 - k_x^2)$ remains real (now negative), the product G_{xx} is **purely imaginary**.

This is confirmed by Figure 1: the real part of G_{xx} (blue solid line) is nonzero only for $k_x/k_0 < 1$, while the imaginary part (red dashed line) is nonzero only for $k_x/k_0 > 1$.

4 Q2: Far Field of a Dipole (4 points)

4.1 Problem Setup

An $\hat{x}$ -directed dipole antenna is centered at the origin with length L along x and width W along y , radiating in free space.

The spatial current distribution is modeled using the open-circuited transmission-line approximation:

$$\vec{J}_x(x, y) = l(x) t(y) \hat{x} \quad (12)$$

with Fourier transform $\tilde{J}_x^{FT}(k_x, k_y) = L(k_x) T(k_y)$, where:

- Transverse distribution:

$$t(y) = \frac{1}{w} \text{rect}\left(\frac{y}{w}\right), \quad T(k_y) = \text{sinc}\left(\frac{k_y w}{2}\right) \quad (13)$$

- Longitudinal distribution:

$$l(x) = \frac{\sin(k_{eq}(\frac{l}{2} - |x|))}{\sin(k_{eq}\frac{l}{2})}, \quad L(k_x) = \frac{2k_{eq}(\cos(\frac{k_x l}{2}) - \cos(\frac{k_{eq} l}{2}))}{(k_{eq}^2 - k_x^2) \sin(k_{eq}\frac{l}{2})} \quad (14)$$

Per the professor's clarification, $k_{eq} = k_0$ (the free-space wavenumber).

4.2 Q2.1: 1D Far-Field Plots in Main Planes

The dipole dimensions are $L = \lambda_0/2 = 5 \text{ mm}$ and $W = \lambda_0/40 = 0.25 \text{ mm}$ at $f = 30 \text{ GHz}$ ($\lambda_0 = 10 \text{ mm}$). The total far field $|\vec{E}^{far}| = \sqrt{|E_\theta|^2 + |E_\phi|^2}$ is computed for $\phi = 0^\circ, 45^\circ$, and 90° .

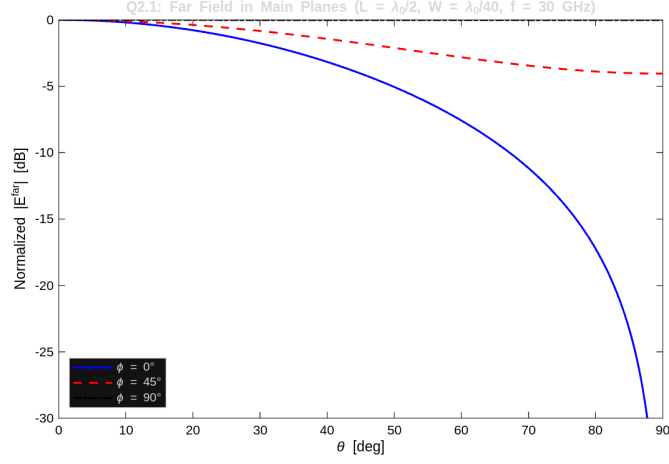


Figure 2: Normalized far-field pattern of the half-wave dipole in the three main planes.

Observations from Figure 2:

- $\phi = 0^\circ$ (**E-plane, xz -plane**): The pattern drops sharply towards $\theta = 90^\circ$ (along the dipole axis), reaching a deep null. This is characteristic of a dipole: no radiation along its own axis.
- $\phi = 90^\circ$ (**H-plane, yz -plane**): The pattern is nearly constant (omnidirectional in the H-plane), as expected for a thin dipole. The slight drop at large θ is due to the finite width W .
- $\phi = 45^\circ$: An intermediate pattern between the E- and H-planes.
- All three planes share the same broadside ($\theta = 0^\circ$) level, confirming the maximum radiation is in the $+z$ direction.

4.3 Q2.2: UV Representation of the Far Field

The far field is plotted in the UV plane ($U = \sin \theta \cos \phi$, $V = \sin \theta \sin \phi$) with a 10 dB dynamic range.

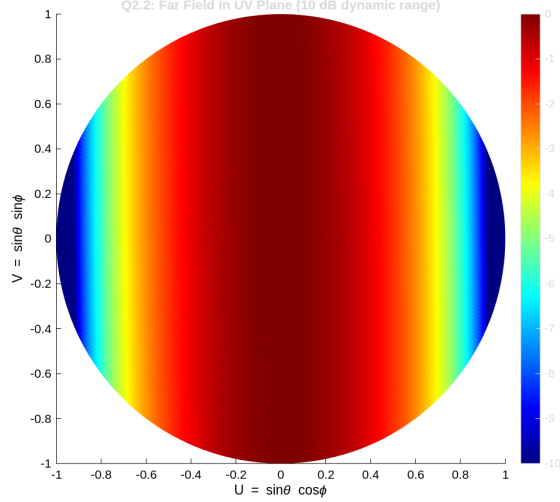


Figure 3: Far-field pattern of the half-wave dipole in the UV plane (10 dB dynamic range). The U -axis corresponds to the dipole axis direction.

The UV representation in Figure 3 clearly shows:

- The maximum radiation is at the center ($U = V = 0$, broadside).
- The pattern exhibits a null along the U -axis ($U = \pm 1, V = 0$), i.e. along the dipole axis ($\hat{x}$ -direction at $\theta = 90^\circ, \phi = 0^\circ$ or 180°).
- The radiation is nearly uniform along the V -axis, consistent with the omnidirectional H-plane pattern.
- The circular boundary ($U^2 + V^2 = 1$) corresponds to $\theta = 90^\circ$ (the horizon).

4.4 Q2.3: Broadside Directivity vs. Frequency

For a fixed dipole with $L = 5$ mm and $W = 0.25$ mm, the broadside directivity is computed as a function of frequency from 5 GHz to 60 GHz. The total radiated power is obtained by integrating over the upper hemisphere and multiplying by 2 (exploiting the symmetry of the free-space pattern about $z = 0$).

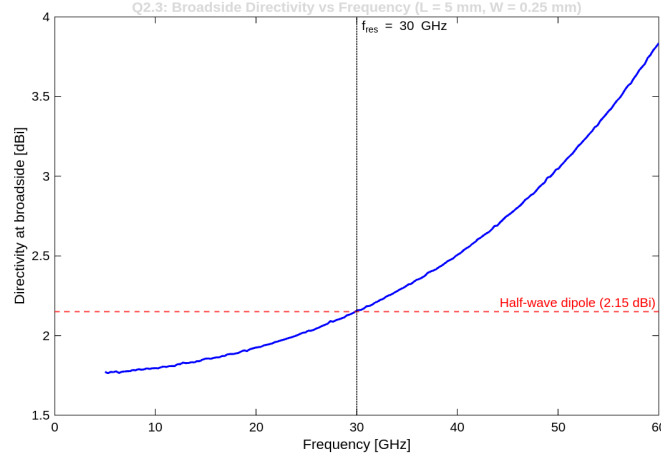


Figure 4: Broadside directivity of the dipole ($L = 5 \text{ mm}$, $W = 0.25 \text{ mm}$) as a function of frequency. The dashed red line marks the theoretical half-wave dipole directivity of 2.15 dBi.

Observations from Figure 4:

- At $f = 30 \text{ GHz}$ (where $L = \lambda_0/2$, i.e. half-wave resonance), the broadside directivity is $\approx 2.12 \text{ dBi}$, matching the theoretical value for a half-wave dipole (2.15 dBi).
- At low frequencies ($f \ll 30 \text{ GHz}$), the dipole is electrically short ($L \ll \lambda$) and approaches the Hertzian dipole limit with $D \approx 1.76 \text{ dBi}$.
- At high frequencies ($f \gg 30 \text{ GHz}$), the dipole becomes electrically long ($L > \lambda$), and the directivity increases as the pattern narrows.

5 Q3: Dipole with a Backing Reflector (3 points)

5.1 Problem Setup

The same dipole ($L = \lambda_0/2$, $W = \lambda_0/40$ at 30 GHz) is now placed at height h above an infinite PEC ground plane. By the image theorem for a perfect electric conductor ($\hat{n} \times \vec{E} = 0$):

- The real source at $z = +h$ has current $+\hat{x}$.
- The image source at $z = -h$ has current $-\hat{x}$ (tangential electric current reverses sign).

The total far field is computed using the free-space SGF applied to both sources. The combined current spectrum includes an array factor:

$$\vec{J}_{\text{total}}(k_{xs}, k_{ys}) = \vec{J}(k_{xs}, k_{ys}) [e^{jk_{zs}h} - e^{-jk_{zs}h}] = \vec{J}(k_{xs}, k_{ys}) \cdot 2j \sin(k_{zs}h) \quad (15)$$

where $k_{zs} = k_0 \cos \theta$ at the stationary phase point.

5.2 Q3.1: Far-Field Comparison with Free-Space Dipole

The far field is computed for $h = 7.5 \text{ mm} = 3\lambda_0/4$ and compared with the free-space dipole pattern from Q2.

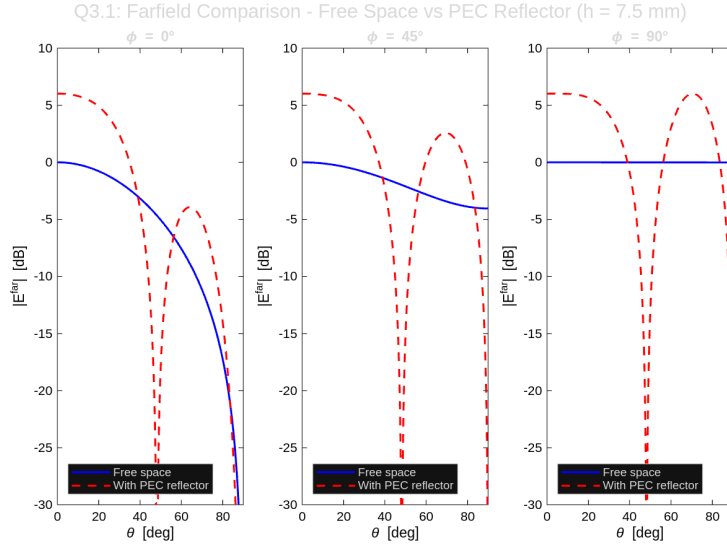


Figure 5: Far-field comparison between the free-space dipole and the dipole with PEC reflector at $h = 7.5 \text{ mm}$, for $\phi = 0^\circ$, 45° , and 90° . Amplitudes are normalized to the free-space broadside level.

Observations from Figure 5:

- The PEC reflector increases the broadside radiation by approximately 6 dB relative to the free-space case, because the image source constructively interferes at broadside when $h = 3\lambda_0/4$ (the array factor $2 \sin(k_0 h \cos \theta)$ evaluated at $\theta = 0$ gives $2 \sin(3\pi/2) = -2$, so $|AF| = 2$, doubling the field and adding 6 dB).
- The reflector introduces additional nulls in the pattern due to destructive interference between the real and image sources at specific angles.

- At $\theta = 90^\circ$ (the horizon), the array factor $2j \sin(k_0 h \cdot 0) = 0$, producing a null for all phi-cuts. This is expected: the PEC ground plane enforces zero tangential electric field at grazing incidence.
- In the H-plane ($\phi = 90^\circ$), the free-space pattern is nearly flat, so the interference lobes from the reflector are clearly visible.

5.3 Q3.3: Broadside Directivity vs. Height h

The broadside directivity (linear scale) is plotted as a function of the distance h from $\lambda_0/10$ to $2\lambda_0$. For the PEC case, the total radiated power is integrated over the upper hemisphere only ($\theta \in [0, \pi/2]$), since no radiation exists below the ground plane.

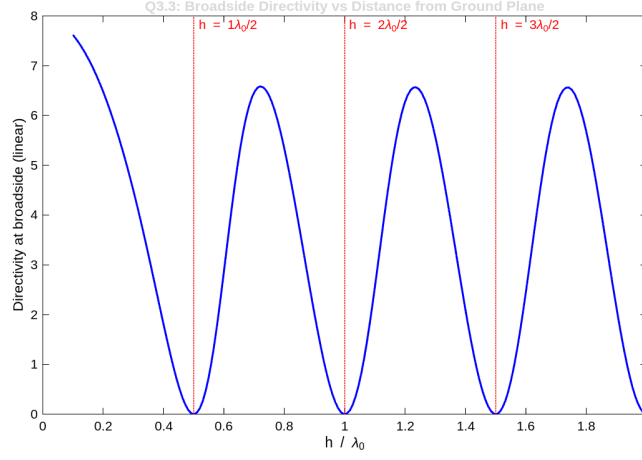


Figure 6: Broadside directivity (linear scale) as a function of the dipole height h/λ_0 above the PEC ground plane.

Observations from Figure 6:

- The directivity oscillates with h , showing clear nulls at $h = n\lambda_0/2$ for $n = 1, 2, 3, \dots$ (marked by red dashed lines). Between the nulls, the directivity peaks at values around 6–7 (linear), corresponding to a significant enhancement over the free-space dipole ($D \approx 1.64$).
- At very small h (near $\lambda_0/10$), the directivity is particularly high (≈ 7.5) because the array factor $2j \sin(k_0 h \cos \theta)$ concentrates radiation near broadside when h is small.

Integration domain for radiated power

Since the image theorem replaces the PEC with image sources, the resulting fields are valid only in the upper half-space ($z > 0$). Therefore, the total radiated power is computed by integrating over the upper hemisphere:

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi/2} U(\theta, \phi) \sin \theta \, d\theta \, d\phi \quad (16)$$

Null at broadside

At broadside ($\theta = 0$), $k_{zs} = k_0$. The array factor is $2j \sin(k_0 h)$, which vanishes when:

$$k_0 h = n\pi \implies h = \frac{n\lambda_0}{2}, \quad n = 1, 2, 3, \dots \quad (17)$$

This is confirmed numerically in Figure 6, where the nulls appear precisely at $h/\lambda_0 = 0.5, 1.0, 1.5$, and 2.0 .

Power radiated for $h = 0$

When $h = 0$, the image source at $z = -h = 0$ coincides spatially with the real source. Since the image current is $-\hat{x}$ (opposite to the real current $+\hat{x}$), the two sources cancel exactly:

$$\text{AF}|_{h=0} = 2j \sin(k_0 \cdot 0) = 0 \quad (18)$$

Therefore the total field is zero everywhere and $P_{\text{rad}} = 0$. This was verified numerically: $P_{\text{rad}}(h = 0) = 0.0 \text{ W}$.